

1st REVIEW

Smart Expense Splitter and Group Budget Management System

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Name of Candidate	Role
Challa Amrutha	Abstract, Introduction, Existing System
Dhurga Meenakshi Yenduri	Problem, Solution(Contribution), Objectives
Earla Ashwith Reddy	Modules, Software requirements, Conclusion and future scope

ABSTRACT

The Smart Expense Splitter and Group Budget Management System is designed to simplify shared expense management through intelligent splitting, budgeting, and financial insights. Unlike traditional expense-splitting systems, it provides predictive budget alerts, spending-pattern analysis, what-if budget simulation, and optimized debt settlement. Users can create groups, record expenses, set flexible sharing rules, and track individual contributions. The system analyzes spending trends to predict budget overruns and identify unusual expenses. It also suggests efficient settlements to reduce unnecessary transactions between members. The proposed system aims to make group financial management ****smarter, faster, transparent, and more proactive****.

INTRODUCTION

Managing shared expenses among friends, families, roommates, or project teams can become difficult when multiple people contribute different amounts. Traditional methods of splitting bills often involve manual calculations, confusion, and difficulty in tracking pending payments. The Smart Expense Splitter and Group Budget Management System is developed to provide a simple and intelligent solution for managing group expenses.

The system allows users to record expenses, automatically calculate individual shares, monitor group budgets, and track payments. It also introduces smart features such as predictive budget alerts, spending-pattern analysis, unusual expense detection, and optimized settlement suggestions. These features help users understand their spending and make better financial decisions. Overall, the system provides a convenient, transparent, and efficient way to manage shared expenses and group budgets.

EXISTING SYSTEM

In the existing system, group expenses are commonly managed using manual calculations, spreadsheets, calculators, or basic expense-splitting applications. Users enter expenses and calculate how much each person owes. These systems mainly focus on recording expenses and dividing bills, but provide limited support for future budgeting and intelligent financial analysis.

Limitations of the existing system:

- Manual calculation can cause errors.
- Limited prediction of future spending.
- No intelligent detection of unusual expenses.
- Basic budget tracking without personalized suggestions.
- Settlement between members may require multiple transactions.
- Limited analysis of group spending patterns.

PROBLEM

Managing shared expenses within groups can be difficult when members contribute different amounts and have different spending needs. Manual calculations and basic expense-splitting systems can lead to errors, confusion, and difficulty tracking budgets. Most existing systems also do not predict future spending or provide intelligent recommendations.

The proposed Smart Expense Splitter and Group Budget Management System aims to solve these problems by providing automatic expense splitting, dynamic budget tracking, spending prediction, unusual-expense detection, and optimized payment settlement. This helps groups manage their finances more accurately, efficiently, and transparently.

SOLUTION(Contribution)

The proposed Smart Expense Splitter and Group Budget Management System provides an intelligent solution for managing shared expenses and group budgets. Its main contributions are:

Smart Expense Splitting: Automatically calculates each member's share based on customized rules.

Dynamic Budget Tracking: Updates the remaining group budget whenever a new expense is added.

Predictive Alerts: Estimates future spending and warns users before the budget is likely to be exceeded.

Spending Analysis: Identifies unusual or rapidly increasing expenses.

Optimized Settlement: Suggests the minimum number of payments needed to settle group debts.

What-If Simulation: Allows users to check how planned expenses may affect the group budget.

Overall, the system makes expense management automated, predictive, transparent, and easier for groups.

Objectives

- To develop a smart and user-friendly system for managing group expenses.
- To automatically calculate and split expenses among group members.
- To provide real-time group budget tracking and remaining-budget updates.
- To predict possible budget overruns and generate timely alerts.
- To analyze spending patterns and identify unusual expenses.
- To reduce unnecessary transactions through optimized debt settlement.
- To help users make better financial decisions through smart spending insights.

Modules

1. User & Group Management Module

User registration and login

Group creation and member management

JWT-based authentication and role management

2. Expense Management Module

Add, edit, and delete expenses

Categorize expenses

Automatic expense splitting

Custom sharing rules

3. Smart Budget Management Module

Set group and category budgets

Real-time budget tracking

Predictive budget-overflow alerts

What-if budget simulation

4. Settlement & Payment Module

Calculate outstanding balances

Optimize settlements between members

Track paid and pending amounts

Minimize unnecessary transactions

Modules

5. Smart Analytics Module

- Spending-pattern analysis
- Unusual expense detection
- Group spending insights
- Personalized budget recommendations

6. Database Management Module

- Relational database for users, groups, expenses, and transactions
- MongoDB for flexible activity/analytics data
- Database indexing, normalization, and optimized queries

7. API & Backend Module

- RESTful APIs using FastAPI
- Pydantic validation
- Database integration using SQLAlchemy/Motor
- API documentation using OpenAPI

8. Microservices & Integration Module

- Separate services for users, expenses, budgets, and settlements
- API Gateway
- Inter-service communication
- Error handling and resilience

9. Deployment & Monitoring Module

- Docker-based deployment
- Service configuration using Docker Compose
- API monitoring and logging
- Prometheus/Grafana-based observability
- CI/CD pipeline integration

SOFTWARE REQUIREMENTS

- **Operating System:** Windows 10/11, Linux, or macOS
- **Frontend:** HTML, CSS, JavaScript / React
- **Backend:** Python with FastAPI
- **Database:** PostgreSQL and MongoDB
- **ORM/Database Tools:** SQLAlchemy, Motor
- **Authentication:** JWT / OAuth2
- **API Documentation:** Swagger / OpenAPI
- **Testing:** Pytest
- **Containerization:** Docker and Docker Compose
- **Development IDE:** Visual Studio Code / PyCharm
- **Version Control:** Git and GitHub
- **API Testing:** Postman
- **Monitoring:** Prometheus and Grafana
- **Optional AI/Smart Features:** Python-based data analysis and prediction libraries

CONCLUSION AND FUTURE SCOPE

The Smart Expense Splitter and Group Budget Management System provides a smart and efficient way to manage shared expenses, budgets, and settlements. It reduces manual calculations through automatic expense splitting, dynamic budget tracking, spending analysis, predictive alerts, and optimized debt settlement. In the future, the system can be enhanced with AI-based financial recommendations, OCR-based receipt scanning, voice expense entry, mobile applications, multi-currency support, and bank/payment integration, making group financial management more automated and convenient.

THANK YOU